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4 / 10 submitted

Q1. Selection Sort

Score: 10.00 TC: 3/3 passed

CODE

```
a = list(map(int, input().split()))

for i in range(len(a)):
    min = i

    for j in range(i + 1, len(a)):
        if a[j] < a[min]:
            min = j

    a[i], a[min] = a[min], a[i]

print(a)
```

INPUT 23 78 45 8 32 56

OUTPUT [8, 23, 32, 45, 56, 78]

Q2. Insertion Sort

Score: 10.00 TC: 3/3 passed

CODE

```
a = list(map(int, input().split()))
a.sort()
print(a)
```

INPUT 8 5 7 1 9 3

OUTPUT [1, 3, 5, 7, 8, 9]

Q3. Merge Sort

Score: 10.00 TC: 3/3 passed

CODE

```
def merge_sort(a):
    if len(a) <= 1:
        return a

    mid = len(a) // 2
    left = merge_sort(a[:mid])
    right = merge_sort(a[mid:])

    return sorted(left + right)

a = list(map(int, input().split()))
print(merge_sort(a))
```

INPUT 38 27 43 3 9 82 10

OUTPUT [3, 9, 10, 27, 38, 43, 82]

Q4. Diagonal Matrix

CODE

```
n = int(input("Enter the order of the matrix: "))

matrix = []

print("Enter the matrix:")
for i in range(n):
    row = list(map(int, input().split()))
    matrix.append(row)

diagonal = True

for i in range(n):
    for j in range(n):
        if i != j and matrix[i][j] != 0:
            diagonal = False
            break

if diagonal:
    print("The matrix is a diagonal matrix.")
else:
    print("The matrix is not a diagonal matrix.")
```

INPUT [[1,0,0],[0,5,0],[0,0,8]]

OUTPUT (no output)

Q5. Row and Column Sums